Mme Doux
11th IB

Quiz 1

Name: .....

Score %	IB Grade

EXERCISE 1

[4 marks]

1. A sequence is defined by $u_1 = 8$, $u_{n+1} = \dfrac{-u_n}{10} + 3$. Write down the first three terms.

2. The general term of a sequence is $u_n = 2n^2 + 1$. Find u_1 and u_5.

3. Is the sequence $80, 40, 20, 10, \ldots$ arithmetic, geometric, or neither? Give the value of d or r.

EXERCISE 2

[6 marks]

1. An arithmetic sequence has first term $u_1 = 12$ and 8th term $u_8 = 47$. Find the common difference d.

2. In an arithmetic sequence, $u_5 = -46$ and $u_{12} = -18$. Find d and u_1.

3. Find the value of m in the arithmetic sequence $m, 15, 3m - 2, \ldots$

EXERCISE 3 **[6 marks]**

1. A geometric sequence has $u_1 = 4$ and common ratio $r = 3$. Write down u_2 and u_3.

2. A geometric sequence has $u_1 = -6$ and $u_4 = -48$. Find the common ratio r.

3. At 9am, a tank contains 800 L of water. Every hour, 25% of the water drains out. Find the amount of water remaining at 1pm, assuming no water is added.

EXERCISE 4 **[2 marks]**

An amount of \$500 is invested in a savings account paying an annual interest rate of 10%, compounded quarterly.

Write down an expression for the amount after 2 years. Do not evaluate it.

THE END